

University of Niagara Falls

Data Analytics MSc



Assignment 1

“Healthcare encounter

Power BI Analysis.”

Data Warehousing/Visualization (CPSC-510-3)

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1. Dataset Description and ETL Summary

The dataset is a healthcare encounters warehouse with one fact table (*Fact Encounter*, 1,500 records) and seven-dimension tables (*Calendar*, *Diagnosis*, *Facility*, *Patient*, *Payer*, *Procedure*, *Provider*), supporting analysis of utilization, cost, and readmissions across facilities, specialties, and time.

All tables were imported, renamed for readability, and profiled using Power Query's Column Quality/Distribution/Profile tools. Two issues were found and corrected:

1. Calendar Table: Quarter column mixed Year and Quarter text (for example, "2019Q1"). It was spitted into a numeric *Quarter Number* column.
2. Fact Encounter: Discharge Date ID used digit-based addition, producing invalid dates (for instance, "Oct 40"). We recalculated with proper *Date.AddDays()* logic so all discharge dates are valid.

All other tables (*Diagnosis*, *Facility*, *Patient*, *Payer*, *Procedure*, *Provider*) were clean. No missing values or duplicate keys were detected. We applied some additional required transformations: Calendar Table gained Start of Year/Quarter/Month, Month Name, and Week of Year fields; Patient Table gained a Patient Profile column (Gender, Age Group, State); Fact Encounter gained a Readmission Category column (Yes/No). All steps are logged in Power Query's Applied Steps pane.

2. Data Model

A star schema was built with Fact Encounter Table at the center, related to all seven-dimension tables via one-to-many relationships on their primary keys. Since Fact Encounter Table has two date keys (Admission/Discharge) both linking to Calendar Table, only the Admission relationship

is active; Discharge is inactive and can be invoked with USERRELATIONSHIP() when needed.

See Appendix A Visualizations for the model diagram [A1](#).

A Matrix visual combining five tables (*Facility*, *Provider*, *Diagnosis*, *Calendar*, *Fact Encounter*) confirmed the model resolves correctly. All subtotals reconciled to the grand total of 1,500 encounters and \$36,294,228 in charges, with no blank values.

3. DAX Measures

Thirteen measures were created, covering Math/Stat, Counting, Logical/Conditional, CALCULATE, and Time Intelligence. The following table explain with detail each measure:

Measure	Category	DAX Logic	Purpose
Average Length of Stay	Math & Stat	AVERAGE(LengthOfStay)	Avg. days per encounter.
Total Billed Charges	Math & Stat	SUM(TotalCharges)	Sum of billed charges.
Total Cost	Math & Stat	SUM(TotalCost)	Sum of actual cost incurred.
Total Encounters	Counting	COUNT(EncounterID)	Total number of encounters.
Distinct Patients	Counting	DISTINCTCOUNT(PatientID)	Unique patient count.
High Cost Encounters	Logical/Conditional	COUNTROWS(FILTER(FactEncounter, TotalCharges>20000))	Counts high-cost encounters.
Readmission Rate %	Logical/Conditional	DIVIDE(SUM(ReadmissionFlag), COUNT(EncounterID), 0)	% of encounters readmitted.
Total Charges Hospital	CALCULATE	CALCULATE(SUM(TotalCharges), FacilityType="Hospital")	Charges filtered to Hospitals.
Avg Cost Readmitted	CALCULATE	CALCULATE(AVERAGE(TotalCost), ReadmissionFlag=1)	Avg. cost, readmitted only.
Average Total Cost	Math & Stat	AVERAGE(TotalCost)	General avg. cost/encounter.
Encounters YTD	Time Intelligence	TOTALYTD(COUNT(EncounterID), Calendar[FullDate])	Running total, year to date.
Charges PY	Time Intelligence	CALCULATE(SUM(TotalCharges), SAMEPERIODLASTYEAR(Calendar[FullDate]))	Charges vs. same period last yr.
Profit Margin	Math & Stat/Logical	DIVIDE([Charges]-[Cost], [Charges], 0)	Profitability % of charges.

Note: Calendar Table was marked as a Date Table (FullDate column) to support the Time

Intelligence measures.

4. Visualization and Key Insights

Three report pages explore different relationships using bar, pie, and line charts, a matrix, a table, and interactive slicers: *Financial & Operational Overview*, *Patient Demographics & Diagnosis Patterns*, and *Readmissions & Quality Insights*. Full-page screenshots of each are provided in Appendix A Visualizations [A2](#), [A3](#) and [A4](#).

Key insights:

1. Orthopedics drives the highest patient volume and billed charges among specialties (\$9.30M of \$36.29M total), a priority area for cost review.
2. Average cost per encounter is nearly identical between Clinics and Hospitals (~\$24,250 vs ~\$24,140) despite different encounter volumes, suggesting facility type is not a strong cost driver.
3. Overall readmission rate is ~15%; high-cost encounters (it is greater than \$130,000) occur in both Clinic and Hospital settings, showing extreme-cost cases are not confined to inpatient care.
4. Average total cost is similar for readmitted vs non-readmitted encounters (~\$13K vs ~\$14K). Readmission alone does not appear to be the main cost driver.
5. Payer mix is well diversified (18.7%–21.4% each across 5 payer types), limiting dependency risk on any single payer.

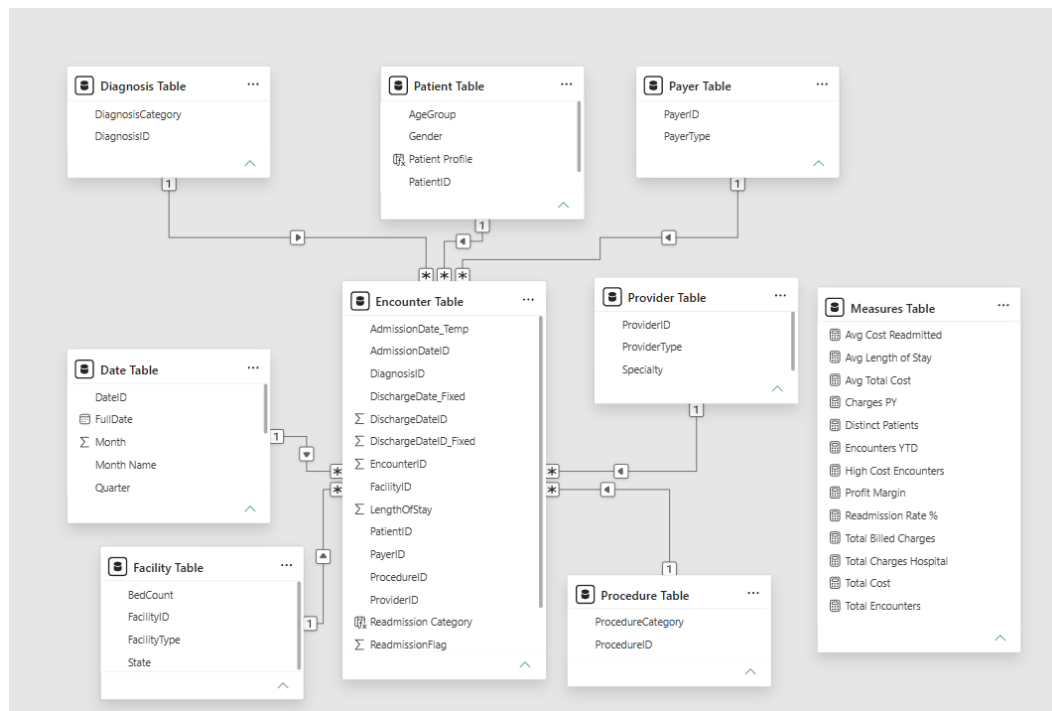
5. Limitations and Optimization

Limitations: the dataset covers a short period (2019–2021) with modest volume (1,500 encounters/500 patients), limiting the robustness of trend analysis; the source system behind the two identified data-quality issues was not available for root-cause review.

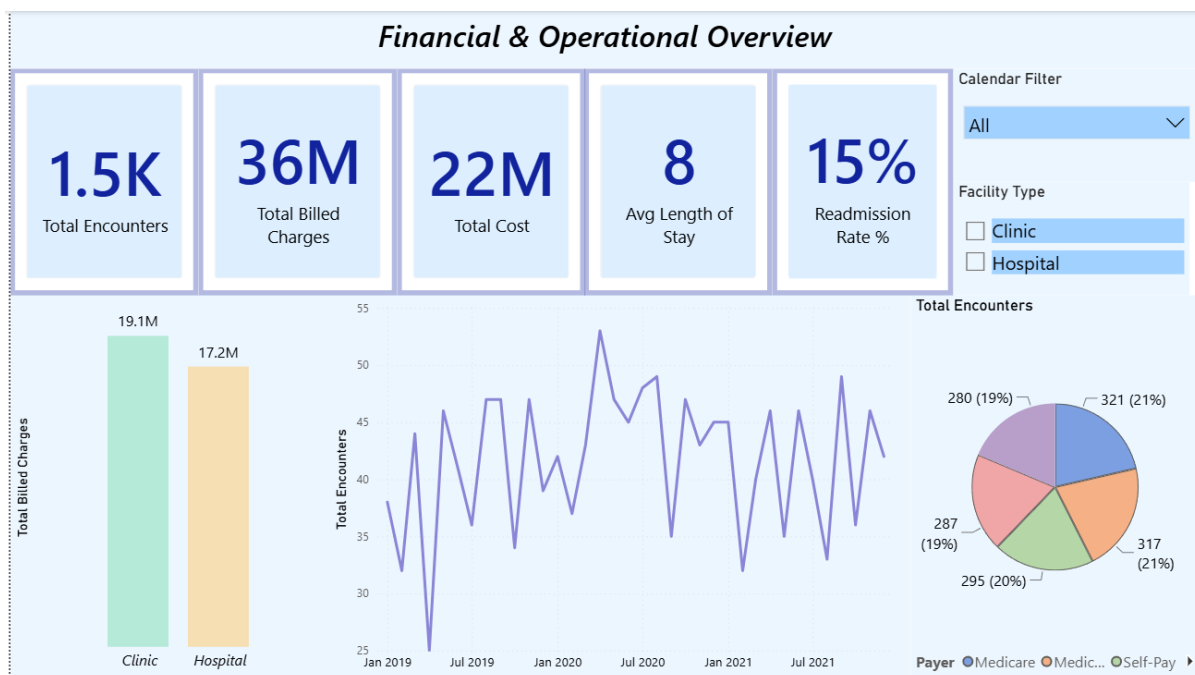
Optimization applied: kept only one active Calendar-to-FactEncounter relationship (Admission Date) rather than duplicating the Calendar Table for Discharge Date, avoiding ambiguous filter propagation while still allowing discharge-based analysis via *USERELATIONSHIP()* when needed.

Appendix A Visualizations

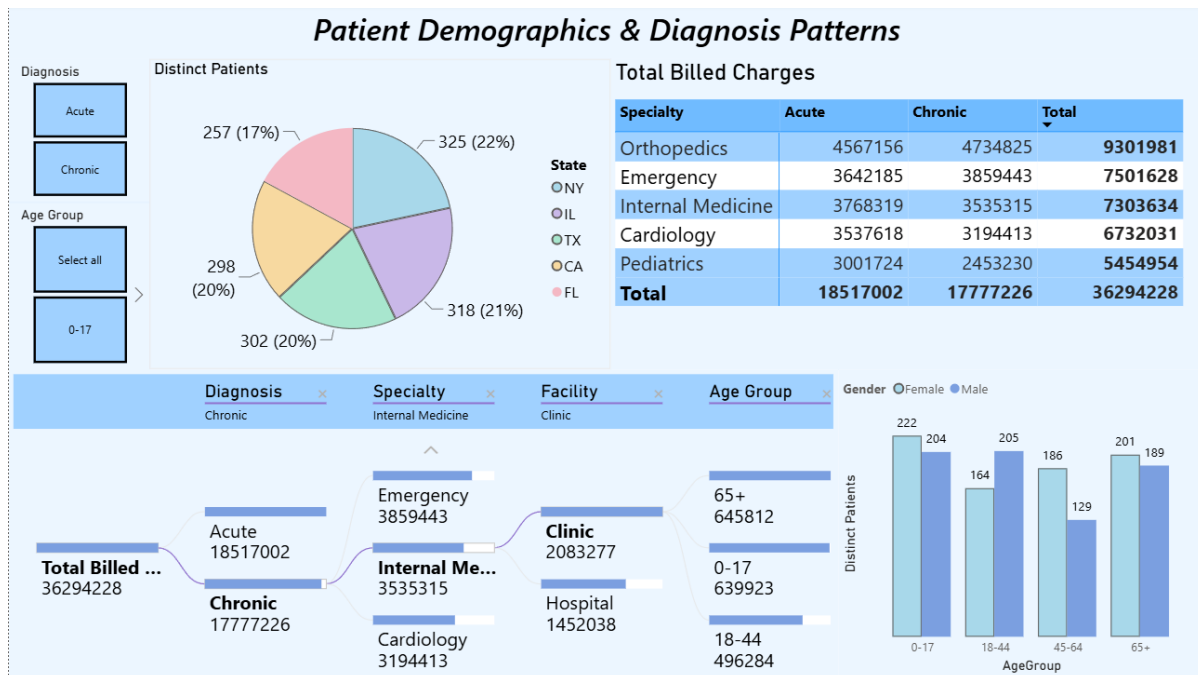
A1Start Diagram



A2 Financial & Operational Overview



A3 Patient Demographics & Diagnosis Patterns



A4 Readmissions & Quality Insights

